END-TO-END SALES ANALYSIS

Sales Performance Dashboard

An end-to-end analysis covering data cleaning, exploratory analysis, business analysis, Excel visualization, and Power BI dashboard development.



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Python | Pandas | Excel | Power BI

DATA • INSIGHTS • DECISIONS

PROJECT OBJECTIVE

The objective of this project is to perform an end-to-end analysis of sales data. The project covers data cleaning and exploratory analysis in Python, business analysis using Pandas, visualization using Excel and Power BI, and the development of actionable business insights and recommendations.

ANALYSIS WORKFLOW

01 — DATA CLEANING

Python + Pandas

02 — EXPLORATORY ANALYSIS

Understand sales, profit and business patterns

03 — BUSINESS ANALYSIS

Analyze country, product, segment, month and discount performance

04 — VISUALIZATION

Excel + Power BI

05 — INSIGHTS & ACTIONS

Convert analysis into business findings and actions

PROJECT DELIVERABLES

PYTHON ANALYSIS
Cleaning & EDA

EXCEL DASHBOARD
Sales visualization

POWER BI DASHBOARD
Interactive analysis

DATASET & DATA CLEANING

DATASET OVERVIEW

700
Records

16
Columns

0
Duplicates After Cleaning

0
Missing Values After Cleaning

DATA CLEANING PERFORMED

01 — Column Names
Removed unnecessary whitespace from column names, including standardizing the Sales column name.

02 — Date Conversion
Converted the Date field to the required date format and checked for invalid dates.

03 — Missing Values
Checked all columns for missing values and handled the missing Discount Band records.

04 — Duplicate Check
Checked the dataset for duplicate records.

05 — Numerical Validation
Checked numerical fields such as Sales, Profit, Units Sold, Discount, and COGS.

06 — Standardization
Standardized the Discount Band values for analysis.

DATA CLEANING ASSUMPTION

53 records had a missing Discount Band and a Discount value of 0. These records were classified as "No Discount" based on the assumption that a zero discount indicates that no discount was applied.

EXPLORATORY DATA ANALYSIS & RESULTS

118.73M	16.89M	1.13M	14.23%
Total Sales	Total Profit	Units Sold	Profit Margin

PERFORMANCE HIGHLIGHTS

01 — COUNTRY PERFORMANCE

USA recorded the highest sales, while France generated the highest profit.

02 — PRODUCT PERFORMANCE

Paseo was the top-performing product by both sales and profit.

03 — SEGMENT PERFORMANCE

Government generated the highest profit, while Enterprise recorded a loss.

TIME & DISCOUNT ANALYSIS

MONTHLY PERFORMANCE

October 2014 recorded the highest monthly sales at 12.38M, while December 2014 recorded the highest monthly profit at 2.03M.

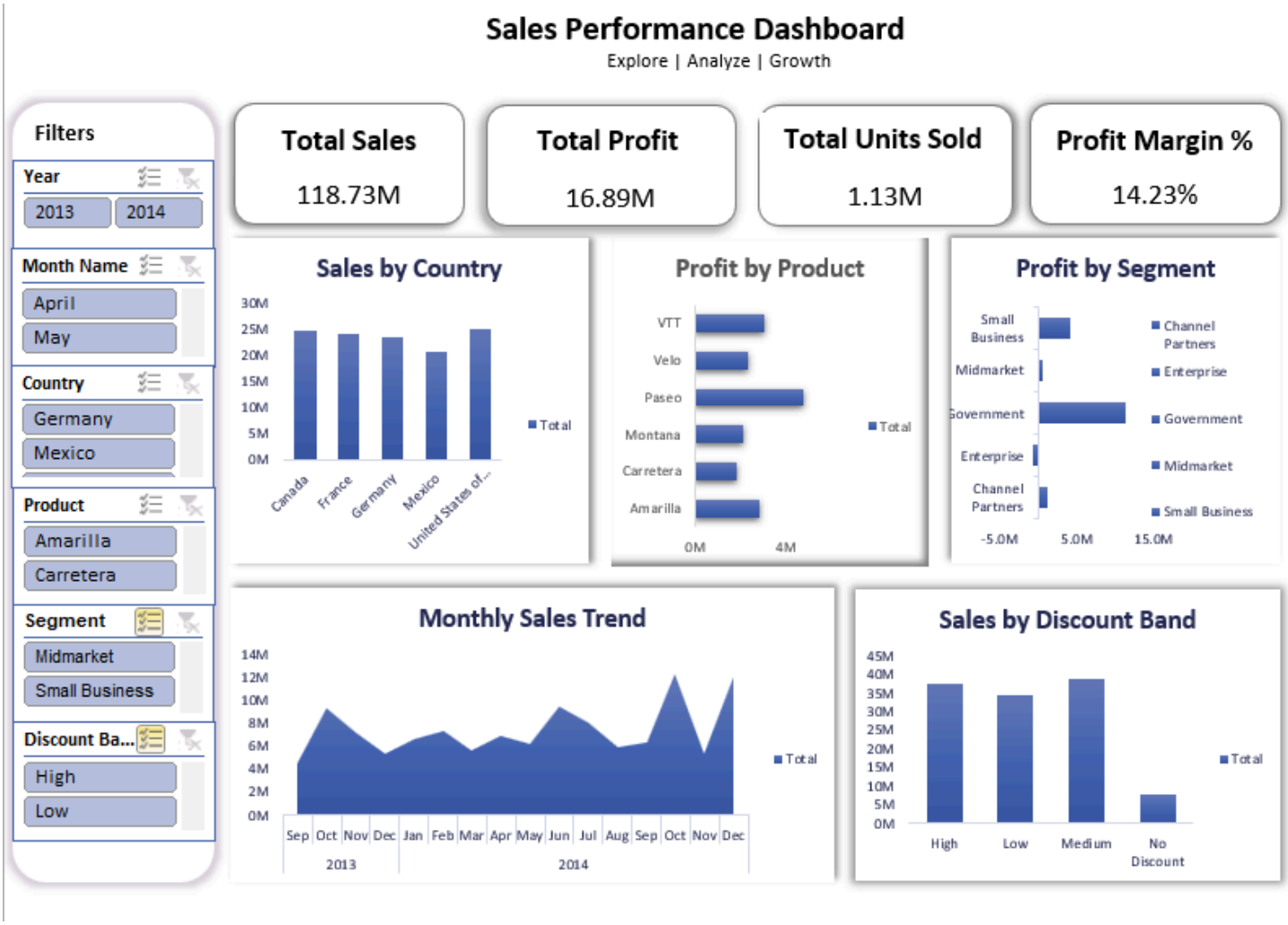
DISCOUNT PERFORMANCE

The Low Discount band generated the highest profit among the assigned discount bands. The High Discount band had the largest total discount amount.

ANALYSIS NOTE

The dataset covers September–December 2013 and January–December 2014, so the year totals should not be interpreted as a full-year growth comparison.

EXCEL DASHBOARD



EXCEL DASHBOARD FEATURES

KPI Overview

Sales, Profit, Units Sold and Profit Margin.

Performance Analysis

Country, Product and Segment performance.

Trend Analysis

Monthly sales and profit patterns.

Interactive Analysis

Filters used to explore different business dimensions.

Purpose: Excel was used to create a business-focused dashboard for summarizing sales performance and identifying important trends.

POWER BI DASHBOARD



POWER BI DASHBOARD FEATURES

Interactive KPI Overview

Total Sales, Total Profit, Units Sold and Profit Margin.

Business Performance Analysis

Country, Product and Segment performance using interactive visuals.

Trend & Discount Analysis

Monthly sales trends and discount-band performance.

Interactive Filters

Year, Month, Country, Product, Segment and Discount Band slicers allow focused analysis.

Purpose: Power BI was used to create an interactive dashboard for exploring sales performance, profitability and business trends.

KEY BUSINESS INSIGHTS

01 — COUNTRY PERFORMANCE

USA generated the highest sales, while France generated the highest profit.

02 — PRODUCT PERFORMANCE

Paseo was the top-performing product by both sales and profit.

04 — TIME PERFORMANCE

October 2014 recorded the highest monthly sales at 12.38M, while December 2014 recorded the highest monthly profit at 2.03M.

03 — SEGMENT PERFORMANCE

Government generated the highest sales and profit. Enterprise recorded a negative profit and needs attention.

05 — DISCOUNT PERFORMANCE

Low Discount generated the highest profit among the assigned discount bands, while High Discount had the largest total discount amount.

Key takeaway: Sales performance is strong overall, but segment profitability and discount effectiveness require closer attention**.**

BUSINESS RECOMMENDATIONS

01 — IMPROVE ENTERPRISE PROFITABILITY

Investigate the Enterprise segment's negative profit and review pricing, costs, discounts and product mix.

02 — LEVERAGE TOP PRODUCTS

Continue focusing on strong-performing products such as Paseo, while evaluating opportunities to improve weaker product performance.

03 — STRENGTHEN HIGH-PROFIT MARKETS

Use the performance of France as a benchmark and identify practices that can improve profitability in other countries.

04 — REVIEW DISCOUNT STRATEGY

Evaluate whether higher discount levels are generating sufficient additional sales and profit. Avoid unnecessary discounting where margins are weak.

05 — USE MONTHLY TRENDS FOR PLANNING

Use high-performing months such as October and December 2014 to support sales planning, inventory decisions and promotional strategies.

Recommended focus: Improve Enterprise profitability, optimize discounting, and build on strong product and market performance.

CONCLUSION & METHODOLOGY

METHODOLOGY

The project followed an end-to-end data analysis workflow, starting with data cleaning and validation using Python and Pandas. The cleaned dataset was then analyzed to identify patterns across countries, products, segments, months and discount bands. Excel and Power BI were used to create visual dashboards and communicate the findings.

KEY TAKEAWAY

Overall, the analysis shows strong sales and profitability performance, with clear differences across countries, products and customer segments. The results also highlight the need to investigate Enterprise losses and evaluate the effectiveness of discounting strategies.

TOOLS USED

PYTHON Data Cleaning & Validation	**PANDAS** Data Analysis
EXCEL Dashboard & Visualization	**POWER BI** Interactive Dashboard

PROJECT OUTCOME

Converted raw sales data into validated analysis, business insights, visual dashboards and actionable recommendations.